

Uncovering the Solar Energy Potential of Reservoirs in Sri Lanka: A Time Series and Machine Learning Approach

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Despite Sri Lanka's high solar energy potential, it contributes only 5% to electricity generation. Land-use conflicts, driven by economic and environmental concerns, hinder large-scale solar deployment. Floating photovoltaic (FPV) systems, mounted on water bodies, offer a solution by saving land and enhancing efficiency through water cooling effects and reduced shading. This study investigates the feasibility of Floating Photovoltaic (FPV) systems on reservoirs, utilising underused water surfaces with minimal land-use conflict. It employs time series and machine learning techniques to predict solar energy output across eighteen major reservoirs in Sri Lanka, offering a sustainable alternative to land-based solar energy expansion. Solar energy generation can be estimated using the equation: $E_{solar} = A \times G_{eff} \times \eta_{eff}$; where, A is the panel area (m^2), G_{eff} is the effective solar radiation after environmental adjustments, and η_{eff} is the efficiency of the solar panel. Effective solar radiation is given by, $G_{eff} = G_{mea} \times (1 - CCo) \times (1 - PCo) \times \frac{P_{ref}}{P_{act}} \times (1 - SAL)$; where G_{mea} is irradiance (W/m^2), CCo is cloud coverage, PCo is precipitation correction, P_{ref} is the reference atmospheric pressure, P_{act} and SAL are the actual surface pressure and surface albedo, respectively. Data were obtained from NASA's Data Access Viewer for the period 2010–2022. Time-series forecasting (SARIMA) and Facebook Prophet models were employed to predict solar energy potential. The solar energy potential for each reservoir was assessed based on a $1 m^2$ area. Both SARIMA and Prophet models revealed strong seasonality in solar energy production. Average daily energy output ranged between 1–4 kWh/ m^2 , with certain months exceeding 5 kWh/ m^2 for specific reservoirs. The highest energy production was typically observed during February, March, and April. To illustrate regional variations in solar energy potential, an Inverse Distance Weighting (IDW) map was generated using the daily average production across all months. The highest solar energy potentials were observed at the *Senanayake Samudraya*, *Lunugamwehera*, *Maduru Oya*, and *Rathkinda-Ulthitiya* reservoirs. Our findings highlight that floating PV is a viable, land-efficient solution to Sri Lanka's energy challenges. It offers a sustainable, scalable method for solar energy production, emphasising the potential of inland water bodies for renewable energy development and supporting national goals for energy security and environmental sustainability.

Keywords: Solar energy, SARIMA, prophet, IDW, reservoirs